

NOTE ON THE ENVELOPE THEOREM

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

1 Introduction

This brief note develops the integral form for the Envelope Theorem that I discussed in class. The treatment follows Milgrom and Segal (2002). We will see that some of the most important results in mechanism design and contract theory are somewhat straightforward applications of the Envelope Theorem. Reading this note is optional, since it requires a significantly higher mathematical background than the rest of the topics in the class. In here I presuppose that not only you have a good course in calculus, but perhaps also a course in Real Analysis. I only recommend going through them if you are really curious or if you are “theoretically inclined.”

Let us first briefly review the usual elementary approach to the Envelope Theorem. In Varian’s (Varian, 1992) classical textbook for example, the argument is as follows.

Let $V(t) = \max_{x \in X} f(x, t)$ for every parameter t in an interval on the real line. Assume that $X \subseteq \mathbb{R}^n$ is nonempty, compact and convex and that f is continuously differentiable in all its arguments. Suppose that $x^*(\cdot)$ maps parameters into optimal choices so that $V(t) = f(x^*(t), t)$. Further suppose that $x^*(t)$ is in the interior of X for every t and that $x^*(\cdot)$ is differentiable.

What is the effect of a change in t on $V(t)$? This question is important because it yields useful comparative statics results, i.e., how the value of a parameterized optimization problem changes when we change one of the exogenous parameters. Differentiation yields

$$V'(t) = f_1(x, t)|_{x=x^*(t)} \frac{dx^*(t)}{dt} + f_2(x, t)|_{x=x^*(t)} = f_2(x^*(t), t).$$

The “intuitive” explanation is that the “ x effect” of a change in t is smoothed out because x is chosen optimally with respect to t . Hence the only effect of a change in t is a “ t effect.”

This reasoning relies on assumptions on X and $f(\cdot, t)$ which are not usually satisfied in mechanism design and contract theory applications.

Example 1.1. Suppose that $X = M$, which is the set of messages sent by an agent to a mechanism. If the mechanism is direct, then $M = T$, the type space. Let μ be a mechanism that maps messages into decisions: for every message m , $\mu(m)$ is the decision taken. The resulting payoff to the agent takes the simple form $u(\mu(m), t)$. If $\sigma(t)$ is the optimal message sent by the agent of type t , then

$$V(t) = \max_m u(\mu(m), t) = u(\mu(\sigma(t)), t).$$

In “applying” the Envelope result presented above, which would imply that $V'(t) = u_2(\mu(\sigma(t)), t)$, several things may go wrong: M need not be a compact and convex subset of $\mathbb{R}^n$, and most importantly, μ need not be differentiable.

2 Preliminaries

To develop the Milgrom-Segal result which will apply under less stringent conditions, we need some preparations. Consider the following definitions.

Definition 2.1. Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f is **continuous at c** if, given any number $\epsilon > 0$, there exists $\delta(c, \epsilon) > 0$ such that if x is any point satisfying $|x - c| < \delta$, then $|f(x) - f(c)| < \epsilon$. If f is continuous at every $c \in A$ we say that f is continuous on A .

Remark. Note that in definition 2.1, the number δ depends on both ϵ and c , i.e., the point at which we are testing for continuity.

Definition 2.2. Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. We say that f is **uniformly continuous** on A if for each $\epsilon > 0$ there is a $\delta(\epsilon) > 0$ such that if $x, u \in A$ are any numbers satisfying $|x - u| < \delta$, then $|f(x) - f(u)| < \epsilon$.

Remark. Note that in definition 2.2, the number δ depends only on ϵ but not on any point of the domain. Thus, it is clear that if f is uniformly continuous, then it is continuous at every point in A .

Exercise 2.1. Prove that the converse is not true, i.e., continuity does not imply uniform continuity by constructing a counter example of a continuous function that is not uniformly continuous.

Remark. If $f : A \rightarrow \mathbb{R}$ is continuous and A is compact, then f is uniformly continuous. This result is called the **Heine-Cantor theorem**. An important special case is when A is a bounded closed interval of real numbers.

Definition 2.3. Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. If there exists a constant $K > 0$ such that

$$|f(x) - f(u)| < K|x - u| \quad (1)$$

for all $x, u \in A$, then f is said to be a **Lipschitz function** (or to satisfy a **Lipschitz condition**) on A .

Remark. The condition (1) that a function $f : I \rightarrow \mathbb{R}$ on an interval I is a Lipschitz function can be interpreted geometrically as follows. If we write the condition as

$$\left| \frac{f(x) - f(u)}{x - u} \right| \leq K, \quad x, u \in I, x \neq u,$$

then the quantity inside the absolute value is the slope of a line segment joining the points $(x, f(x))$ and $(u, f(u))$. Thus a function f satisfies a Lipschitz condition if and only if the slopes of all line segments joining two points on the graph of $y = f(x)$ over I are bounded by some number K .

Exercise 2.2. Show that if $f : A \rightarrow \mathbb{R}$ is a Lipschitz function, then f is uniformly continuous on A .

Definition 2.4. A function $f : [a, b] \rightarrow \mathbb{R}$ is **absolutely continuous** if for every $\epsilon > 0$, there exists $\delta(\epsilon) > 0$ such that for any finite set of disjoint intervals $[a_1, b_1], \dots, [a_n, b_n]$ in $[a, b]$

$$\sum_{j=1}^n (b_j - a_j) < \delta \Rightarrow \sum_{j=1}^n |f(b_j) - f(a_j)| < \epsilon.$$

Remark. Note that in the definition of absolute continuity 2.4, the number δ only depends on ϵ but it cannot depend on the intervals $[a_1, b_1], \dots, [a_n, b_n]$ or the number n of intervals.

Several important important results follow:

Claim 2.3. If f is absolutely continuous, then f is continuous. There are continuous functions which are not absolutely continuous.

Exercise 2.4. Prove the first part of claim 2.3, i.e., that if f is absolutely continuous, then it is continuous.

Claim 2.5. If f satisfies a Lipschitz condition $|f(x) - f(y)| < C|x - y|$ for every $x, y \in [a, b]$, then f is absolutely continuous.

Exercise 2.6. Prove claim 2.5

Claim 2.7. If f is differentiable and f' is bounded, then f is absolutely continuous, since by the Mean Value Theorem, $|f(b_j) - f(a_j)| \leq \sup_x |f'(x)|(b_j - a_j)$. This implies, in particular, that continuously differentiable functions on an interval are absolutely continuous.

Claim 2.8. If f is differentiable and monotone, then f is absolutely continuous. (Koliha, 2006)

Claim 2.9. If f is convex, then f is absolutely continuous.

We will also need some notions of convergence for sequences of functions.

Definition 2.5. A sequence of (f_n) of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ **converges pointwise** to a function $f : A_0 \rightarrow \mathbb{R}$ on $A_0 \subseteq A$ if for each ϵ and each $x \in A_0$ there is a natural number $K(\epsilon, x)$ such that if $n \geq K(\epsilon, x)$ then

$$|f_n(x) - f(x)| < \epsilon$$

Definition 2.6. A sequence of (f_n) of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ **converges uniformly** on $A_0 \subseteq A$ to a function $f : A_0 \rightarrow \mathbb{R}$ if for each ϵ there is a natural number $K(\epsilon)$ such that if $n \geq K(\epsilon)$ then

$$|f_n(x) - f(x)| < \epsilon \quad \text{for all } x \in A_0.$$

Remark. Note that in definition 2.6, the number K depends only on ϵ but not on x . Thus, it is clear that if the sequence (f_n) converges uniformly to f then it also converges pointwise.

Exercise 2.10. Prove that the converse is not true, i.e., that pointwise convergence does not imply uniform convergence by constructing an example of a sequence of functions that converge pointwise but not uniformly.

3 An example of a continuous function which is not absolutely continuous

This section proves the second part of claim 2.3, i.e., that there are continuous functions that are not absolutely continuous. I prove this by going over the construction of a famous counterexample, the **Cantor-Lebesgue function**, $f : [0, 1] \rightarrow \mathbb{R}$, which is continuous but not absolutely so. Feel free to skip this part on a first reading.

We need more preparations.

Definition 3.1. A set C is **dense in itself** if for every $x \in C$ and every $\epsilon > 0$, there exists $x' \in C$ such that $0 < |x - x'| < \epsilon$. In other words, C is dense in itself if every point of C is a *limit point* of C .

Definition 3.2. A **perfect set** is a closed set which is dense in itself.

Claim 3.1. A perfect set is uncountable.

Definition 3.3. If I is an interval, let $v(I)$ denote its **volume**, which is simply the difference between its endpoints.

Definition 3.4. Let $E \subset \mathbb{R}$. Then, the **Lebesgue outer measure** is constructed as follows. Let S be a countable collection of intervals I_k which cover E . Define $\sigma(S) := \sum_{I_k \in S} v(I_k)$. The **Lebesgue outer measure** of E , denoted $|E|_e$ is defined as $|E|_e = \inf \sigma(S)$.

Remark. Note that $|E|_e$ always exists as it is the infimum of nonnegative numbers.

Definition 3.5. A set $E \subseteq \mathbb{R}$ is **Lebesgue measurable** if for all $\epsilon > 0$, there exist an open set G containing E such that $|G \setminus E|_e < \epsilon$.

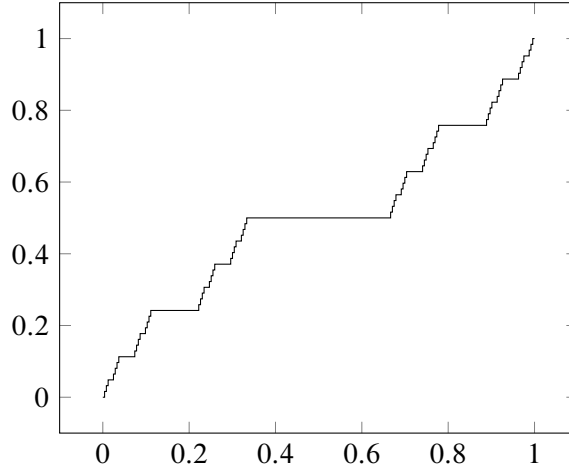


Figure 1: Cantor-Lebesgue function

Definition 3.6. If E is measurable, its **Lebesgue measure** is given by $|E| = |E|_e$.

Remark. Using the Axiom of Choice, one can construct non-measurable subsets of $\mathbb{R}$. The classical example is the *Vitali set*.

Now let us construct the **Cantor set**. Take $[0, 1]$ and consider the following procedure of consecutive elimination of open intervals from it.

Step 1: Remove $(1/3, 2/3)$ so that $C_1 = [0, 1/3] \cup [2/3, 1]$ remains.

Step 2: Remove $(1/9, 2/9) \cup (7/9, 8/9)$ so that $C_2 = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1]$ remains.

... *ad infinitum* ...

The **Cantor set** C is defined as $C = \bigcap_{k=1}^{\infty} C_k$.

At each stage k , a set C_k which is a union of 2^k closed intervals remains. Each of the intervals that make up C_k has a volume of $\frac{1}{3^k}$. C contains the end points of each of these intervals. If $x \in C$, then $x \in C_k$ for every k and therefore x is a limit point of end points of intervals. Hence C is perfect. But its outer measure is zero since for every k , $C \subseteq C_k$ implying that $|C|_e \leq |C_k|_e$ and $|C_k|_e < \frac{2^k}{3^k} \rightarrow 0$.

We are now in position to define the **Cantor-Lebesgue function**. Let $D_k = [0, 1] \setminus C_k$ so that $D_k = \bigcup_{j=1}^{2^k-1} I_j^k$ is the union of open intervals (ordered from left to right) removed at the end of the first k stages. Define a sequence of continuous nondecreasing functions $\{f_k\}$ where $f_k : [0, 1] \rightarrow \mathbb{R}$ is defined as follows. $f_k(0) = 0$, $f_k(1) = 1$, $f_k(x) = \frac{j}{2^k}$ if $x \in I_j^k$, $j = 1, \dots, 2^k-1$ and f_k is linear on C_k . Note that $f_{k+1} = f_k$ on I_j^k , $j = 1, \dots, 2^k-1$. Also note that $|f_k(x) - f_{k+1}(x)| < \frac{1}{2^k}$. Hence $\sum_{\ell=j}^k (f_\ell - f_{\ell+1})$ converges uniformly on $[0, 1]$ and therefore $\{f_k\}$ converges uniformly on $[0, 1]$. Let $f = \lim_{k \rightarrow \infty} f_k$. Then f is uniformly continuous and nondecreasing. Moreover, $f(0) = 0$ and $f(1) = 1$. The function f is constant on every interval removed in the construction of C . The function f is called the Cantor-Lebesgue function. It provides an example of a continuous function which is not absolutely continuous. Figure 1 shows the graph of the Cantor-Lebesgue function.¹ Recall that if $C_k = \bigcup_{j=1}^{2^k} [a_j^k, b_j^k]$ is the set that remains in stage k , then $|C_k| \rightarrow 0$ but $\sum_{j=1}^{2^k} |f(b_j^k) - f(a_j^k)| = 1$.

¹To be very precise, figure 1 shows the fifth step in the iterative construction of the Cantor-Lebesgue function above, i.e. the step f_5 . The true Cantor-Lebesgue function was constructed by taking the limit of the sequence $\{f_k\}$. However, f_5 provides an accurate illustration of how the function “looks like.”

4 The envelope theorem

We will need the following theorem as a stepping stone.

Theorem 4.1 (Fundamental Theorem of Calculus for Lebesgue Integrals). *If $-\infty < a < b < \infty$ and $f : [a, b] \rightarrow \mathbb{R}$, then the following are equivalent*

1. f is absolutely continuous.
2. There is a Lebesgue integrable function $g : [a, b] \rightarrow \mathbb{R}$ such that $f(x) - f(a) = \int_a^x g(s)ds$.
3. f is differentiable almost everywhere on $[a, b]$, f' is Lebesgue integrable on $[a, b]$ and $\int_a^x f'(s)ds = f(x) - f(a)$.

Example 4.1. *Hotelling's Lemma.* Let $\pi(p) = \max_{y \in Y} p \cdot y$ and $y^*(p) \in \arg \max_{y \in Y} p \cdot y$. One can show that:

1. If $\pi(\cdot, p_{-j})$ is absolutely continuous, then for every $\hat{p}_j$, $\pi(\hat{p}_j, p_{-j}) = \pi(0, p_{-j}) + \int_0^{\hat{p}_j} y_j^*(z, p_{-j})dz$.
2. If $\pi(\cdot, p_{-j})$ is differentiable at $\hat{p}_j$, then $y_j^*(\hat{p}_j, p_{-j}) = \frac{\partial}{\partial p_j} \pi(\hat{p}_j, p_{-j})$.

Note that the first conclusion depends only on the absolute continuity of π . It does not require that the production set Y be convex or that y_j^* be differentiable or continuous, or even that it exists everywhere.

The family of problems will be parametrized by $t \in [0, 1]$. Consider the related functions

$$\begin{aligned} V(t) &= \sup_{x \in X} f(x, t), \\ X^*(t) &= \{x \in X : f(x, t) = V(t)\}, \\ x^*(t) &\in X^*(t). \end{aligned}$$

The function V is the value function, or the envelope function. If for each x , we plot $f(x, \cdot)$, V is the upper envelope of these functions. X^* gives the set of optimal solutions to $\sup_{x \in X} f(x, t)$. For some t , $X^*(t)$ may be empty. Any function x^* is a selection from X^* if $x^*(t) \in X^*(t)$ for every t such that $X^*(t)$ is non-empty.

Envelope theorems report a relation between V and any selection x^* from X^* .

Theorem 4.2 ((Milgrom and Segal, 2002) Envelope Theorem). *Suppose that*

1. $f(x, \cdot)$ is differentiable almost everywhere, its derivative $f_2(x, \cdot)$ is integrable and $f(x, t) = f(x, 0) + \int_0^t f_2(x, s)ds$.
2. $|f_2(x, t)| < c(t)$ for all x and almost all t for some integrable function $c : [0, 1] \rightarrow \mathbb{R}$.
3. $X^*(t)$ is nonempty for almost all t .

Then, for any selection x^* from X^* ,

$$V(t) = f(x^*(t), t) = f(x^*(0), 0) + \int_0^t f_2(x^*(s), s)ds$$

Proof. The proof proceeds in 2 steps. In step 1 we show that V is absolutely continuous. In step 2, we show that if x^* is a selection from X^* , then $V'(t) = f_2(x^*(t), t)$ for almost every t . By the fundamental theorem of calculus, (theorem 4.1), these two steps will imply the desired equality.

Step 1: For every t' and t ,

$$\begin{aligned}
|V(t) - V(t')| &\leq \sup_{x \in X} |f(x, t) - f(x, t')| \\
&= \sup_{x \in X} \left| \int_{t'}^t f_2(x, s) ds \right| \\
&\leq \int_{t'}^t \sup_{x \in X} |f_2(x, s)| ds \\
&\leq \int_{t'}^t c(s) ds
\end{aligned}$$

To see the first inequality note that

$$|V(t) - V(t')| = \left| \sup_x f(x, t) - \sup_x f(x, t') \right|.$$

There are two cases to consider:

$$\begin{aligned}
\sup_x f(x, t) - \sup_x f(x, t') &\geq 0 \Rightarrow \\
f(x^*(t), t) - f(x^*(t), t') &\geq \sup_x f(x, t) - \sup_x f(x, t') \geq 0 \Rightarrow \\
\left| \sup_x f(x, t) - \sup_x f(x, t') \right| &\leq |f(x^*(t), t) - f(x^*(t), t')| \leq \sup_x |f(x, t) - f(x, t')|
\end{aligned}$$

and

$$\begin{aligned}
\sup_x f(x, t) - \sup_x f(x, t') &\leq 0 \Rightarrow \\
f(x^*(t'), t) - f(x^*(t'), t') &\leq \sup_x f(x, t) - \sup_x f(x, t') \leq 0 \\
\left| \sup_x f(x, t) - \sup_x f(x, t') \right| &\leq |f(x^*(t'), t) - f(x^*(t'), t')| \leq \sup_x |f(x, t) - f(x, t')|.
\end{aligned}$$

To see the second inequality, we first establish that $|f f| \leq f|f|$. Define functions $f^+ = \max(f, 0)$ and $f^- = -\min(f, 0)$ so that $f = f^+ - f^-$ and $|f| = f^+ + f^-$. Note that f^+ and f^- are nonnegative valued functions. Recall that if a and b are nonnegative, then $|a - b| \leq a + b$. It follows that $|f f| = |f f^+ - f f^-| \leq f(f^+ + f^-) = f|f|$. Now we use this inequality to get for every x

$$\left| \int f_2(x, s) ds \right| \leq \int |f_2(x, s)| ds \leq \int \sup_x |f(x, s)| ds$$

implying that $\sup_x \left| \int f_2(x, s) ds \right| \leq \int \sup_x |f_2(x, s)| ds$

Now fix $\epsilon > 0$. Note that c is positive valued almost everywhere. Since c is also integrable, there is some positive number M such that

$$\int_{\{c(s) > M\}} c(s) ds < \frac{\epsilon}{2}.$$

Let $\delta < \frac{\epsilon}{2M}$ and select non-overlapping intervals $[a_1, b_1], \dots, [a_n, b_n]$ (from left right, i.e., $b_k < a_{k+1}$) in $[0, 1]$. If $\sum_{i=1}^n (b_i - a_i) < \delta$, then

$$\begin{aligned}
\sum_{i=1}^n |V(b_i) - V(a_i)| &\leq \sum_{i=1}^n \int_{a_i}^{b_i} c(s) ds \\
&\leq \sum_{i=1}^n \int_{a_i}^{b_i} c(s) ds \\
&\leq \sum_{i=1}^n \int_{\{c(s) > M\}} c(s) ds + \sum_{i=1}^n \int_{\{c(s) > M\} \cap [a_i, b_i]} c(s) ds \\
&< \frac{\epsilon}{2} + M \sum_{i=1}^n (b_i - a_i) \\
&< \epsilon
\end{aligned}$$

and V is absolutely continuous. The third inequality follows because c is nonnegative valued almost everywhere (hypothesis 2 in the theorem) and the area of integration expands to:

$$\bigcup_{i=1}^n [a_i, b_i] \subseteq \{s : c(s) > M\} \cup \left(\{s : c(s) \leq M\} \cap \bigcup_{i=1}^n [a_i, b_i] \right)$$

Step 2: Since V is absolutely continuous, it is differentiable almost everywhere. Let t be a point of differentiability. Then right and left derivatives at t exist and are the same. Let $x^* \in X^*(t)$ so that $V(t) = f(x^*, t)$. Since for every Δ and $V(t + \Delta) \geq f(x, t + \Delta)$, it follows that

$$V(t + \Delta) - V(t) \geq f(x^*, t + \Delta) - f(x^*, t)$$

For every $\Delta > 0$ we have

$$\frac{V(t + \Delta) - V(t)}{\Delta} \geq \frac{f(x^*, t + \Delta) - f(x^*, t)}{\Delta}$$

and if $\Delta \searrow 0$, we get $V'(t) \geq f_2(x^*(t), t)$. Similarly, for every $\Delta < 0$, we have

$$\frac{V(t + \Delta) - V(t)}{\Delta} \leq \frac{f(x^*, t + \Delta) - f(x^*, t)}{\Delta}$$

and if $\Delta \nearrow 0$ we get $V'(t) \leq f_2(x^*(t), t)$. Hence $V'(t) = f_2(x^*(t), t)$ and the proof is complete. $\square$

5 Application to Mechanism Design and Contract Theory with a Single Agent

Let us apply the Envelope theorem in a mechanism design/contracting setting. Consider, for simplicity, a single agent mechanism design problem with $A \subseteq \mathbb{R}$, $T = [0, 1]$ and $v(a, t) = at$. Let (ψ, x) be a mechanism. Consider for every t , the following parametrized maximization problem:

$$V(t) = \max_{t' \in T} \psi(t')t - x(t').$$

Note that IC is the following requirement: $V(t) = \psi(t)t - x(t)$. Using the Envelope Theorem, we can get $V(t) = V(0) + \int_0^t \psi(s) ds$. Note that IC implies

$$\begin{aligned}
\psi(t)t - x(t) &= V(t) \\
&= V(0) + \int_0^t \psi(s) ds \\
&= -x(0) + \int_0^t \psi(s) ds
\end{aligned}$$

giving the formula

$$x(t) = x(0) + \psi(t)t - \int_0^t \psi(s)ds$$

Note now that if (ψ, x) and $(\hat{\psi}, \hat{x})$ are two IC mechanism and if $\psi = \hat{\psi}$, then for every t , $x(t) - \hat{x}(t)$ is equal to the constant $x(0) - \hat{x}(0)$. This observation is generalizable to multi-agent settings and is one of the fundamental workhorses of mechanism design theory as well as the theory of multi-agent contracts and auctions. It is usually referred to as the revenue (or payoff) equivalence principle.

In the current simple setting, the Envelope Theorem can be obtained using more direct arguments, which may be useful in understanding the exact nature of incentive compatibility. Note that for any (ψ, x)

$$V(t) = \max_{t'} \psi(t')t - x(t') \quad \text{for every } t$$

implying that V is a maximum of affine functions. Consequently, V is absolutely continuous and the integral of its derivative. Moreover V is differentiable almost everywhere. Now suppose that (ψ, x) is IC. We can use this to calculate the derivative of V whenever it exists, as follows. IC implies that for every t and t'

$$V(t') = \psi(t')t' - x(t') \geq \psi(t)t' - x(t) = V(t) + \psi(t)(t' - t)$$

This means that for every t' , $\psi(t')$ is the slope of a function which supports V . Hence, if t is a point of differentiability, then $V'(t) = \psi(t)$. As a consequence, ψ is nondecreasing, and by the FTC, $V(t) = V(0) + \int_0^t \psi(s)ds$.

References

- KOLIHA, J. J. (2006): “A fundamental theorem of calculus for Lebesgue integration,” *The American Mathematical Monthly*, 113, 551–555.
- MILGROM, P. AND I. SEGAL (2002): “Envelope theorems for arbitrary choice sets,” *Econometrica*, 70, 583–601.
- VARIAN, H. R. (1992): *Microeconomic analysis*, Norton New York.